THE FAMILY {U -> X, Y -> X} OF THE FORMAL GLUEING SITUATION IS REFINED BY AN FPQC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2271-2276 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2278-2299, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_spaces-pushouts.json (pb=1233, stmt 2271-2276, proof 2278-2299).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 096V: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2299–2299. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AEV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2046–2060. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03L7: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2299–2299. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 022A: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2299–2299. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0AEV there exists an fpqc covering $\{X_i \to X\}_{i \in I}$ refining the family $\{U \to X, Y \to X\}$.*

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. For the definition and general properties of fpqc coverings we refer to Topologies, Section 022A. In particular, we can first choose an étale covering $\{X_i \to X\}$ with X_i affine and by base changing Y, Z, and U to each X_i we reduce to the case where X is affine. In this case U is quasi-compact and hence a finite union $U = U_1 \cup \ldots \cup U_n$ of affine opens. Then Z is quasi-compact hence also $f^{-1}Z$ is quasi-compact. Thus we can choose an affine scheme W and an étale morphism $h : W \to Y$ such that $h^{-1}f^{-1}Z \to f^{-1}Z$ is surjective. Say $W = \mathrm{Spec}(B)$ and $h^{-1}f^{-1}Z = V(J)$ where $J \subset B$ is an ideal of finite type. By Pro-étale Cohomology, Lemma 096V there exists a localization $B \to B'$ such that points of $\mathrm{Spec}(B')$ correspond exactly to points of $W = \mathrm{Spec}(B)$ specializing to $h^{-1}f^{-1}Z = V(J)$. It follows that the composition $\mathrm{Spec}(B') \to \mathrm{Spec}(B) = W \to Y \to X$ is flat as by assumption $f : Y \to X$ is flat at all the points of $f^{-1}Z$. Then $\{\mathrm{Spec}(B') \to X, U_1 \to X, \ldots, U_n \to X\}$ is an fpqc covering by Topologies, Lemma 03L7. $\square$